Heah Hung Xun

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Summary

Second-year B.Sc. Physics undergraduate at **Xiamen University Malaysia** with interests in **quantum foundations, mathematical physics, quantum field theory, and physics beyond the Standard Model**. Current work focuses on the role of complex numbers in quantum theory, the quantum–classical interface, and connections between quantum mechanics and statistical mechanics. I combine theoretical study with scientific computing in Python and C++, while developing proficiency in ROOT and GEANT4.

Research & Technical Projects

Phase-Space Quantum Mechanics & Semiclassical Limits *2026/01*

- Authored a self-contained pedagogical monograph on quantum mechanics formulated in phase space.
- Developed the Weyl–Wigner correspondence mapping operators to phase-space symbols.
- Derived the Moyal $\star$-product and Moyal bracket, showing explicitly how classical Poisson dynamics emerge in the semiclassical limit $\hbar \to 0$.
- Analysed the quantum–classical transition via coarse-graining, interpreting classicality as information loss.
- Closely connected to deformation quantization and semiclassical analysis.

Angular Momentum Algebra & Hydrogen Atom in 3D *2025/12*

- Developed a rigorous algebraic treatment of angular momentum in quantum mechanics.
- Emphasised $\mathfrak{so}(3)$ and $\mathfrak{su}(2)$ structure, ladder operators, and representation theory.
- Applied spin and orbital angular momentum formalism to the hydrogen atom without relying on heuristic arguments.
- Focused on structural motivation rather than postulated rules.

Quantum Random Walks: Theory and Developments *2025/04 – 2025/09*

- Systematic survey of discrete- and continuous-time quantum walks from a mathematical and physical perspective.
- Derived classical diffusion limits and contrasted them with quantum ballistic spreading ($\sigma \sim t$).
- Reviewed algorithmic constructions (SKW, Szegedy) and open problems in mixing and state transfer.
- Implemented and visualised quantum walk dynamics numerically using Python (NumPy, QuTiP, Matplotlib).

Current Research Directions & Training

- **Real formulations of quantum mechanics:** On the necessity of complex numbers in Quantum Mechanics, and studying recent real-number reformulations and their physical interpretation.
- **Quantum–classical and statistical structure:** Critically reading an ongoing exchange on reconstructing quantum waves from classical action, including the roles of propagated densities, Madelung variables, and the Bohm quantum potential [2405.06328, 2605.02621, 2605.20443].
- **Quantum field theory and BSM physics:** Systematically studying canonical quantization, free fields, Fock space, propagators, and introductory perturbation theory (Attending a seminar series by Prof. Yi-Zen Chu), with longer-term interest in Higgs-portal models and other beyond-the-Standard-Model scenarios.
- **Foundations of physical theories:** Incoming participant in the 2026 virtual Summer School

on the Assumptions of Physics, focused on deriving physical theories from explicit operational and mathematical assumptions.

Education

2023 – present	**B.Sc. in Physics**, Xiamen University Malaysia GPA: 3.75 / 4.00 *Selected Coursework:* Quantum Mechanics I, Mathematical Methods, Theoretical Mechanics, Linear Algebra, Differential Equations, Electric Circuits, Electrodynamics, Numerical Methods, Optics

Leadership & Activities

2025/09 – present **Vice President, Physics Student Council** — Coordinate academic events, peer mentoring, and student engagement.

- Delivered departmental talk *A Year in Physics* [slides]; preparing upcoming talk on LaTeX and Git.
- Founded and host the department Physics Forum, running a student journal club.

2025/09 – present **Vice President, Astronomy Club** — Organise telescope sessions and member-led knowledge sharing.

Skills

Mathematical Physics	Hilbert space formalism, operator algebras, angular momentum theory, $\mathfrak{su}(2)$ representations, semiclassical limits, deformation quantization, phase-space methods
Quantum Topics	Quantum foundations, real and complex formulations of quantum mechanics, quantum walks, quantum dynamics, Wigner functions, Moyal $\star$-product, Weyl quantization, Madelung–Bohm formulation
QFT & Particle Physics	Systematic study of canonical quantization, free scalar fields, Fock space, propagators, and Feynman diagrams; interests in Higgs-portal and other BSM models, together with detector concepts and muon physics
HEP Software	ROOT (actively learning — histograms, trees, fitting), GEANT4 (familiarising — geometry, physics lists, detector simulation), C++ (learning, primary language for ROOT / GEANT4 workflows)
Machine Learning / AI	Numpy-based model building, familiarity with scikit-learn; interest in ML applications to HEP (trigger, classification, anomaly detection)
Scientific Computing	Python (NumPy, SciPy, QuTiP, Matplotlib, Pandas), MATLAB, LaTeX, Git, Jupyter
Tools & Environment	Linux, VS Code, GitHub, Vim, Emacs